Student Performance Worksheet & Mark List

Computer Applications Practical Examination | Question 4 Response (16 Marks)

1. Complete Student Mark List Worksheet

The table below presents the generated worksheet containing all subject marks, calculated total scores out of 800, percentage calculations, and division awards based on predefined conditional logic:

Student Name	Hindi (out of 200)	Maths (out of 200)	Finance (out of 400)	Total (T) (out of 800)	Percentage (P) (%)	Division (D)
Arvind	150	150	300	**600**	**75.00%**	1st with Distinction
Ajay	165	145	320	**630**	**78.75%**	1st with Distinction
Rehman	160	136	250	**546**	**68.25%**	1st Division
Roger	145	149	190	**484**	**60.50%**	1st Division
Samay	135	154	250	**539**	**67.38%**	1st Division
Baljeet	130	165	350	**645**	**80.63%**	1st with Distinction
Class Average	**147.50**	**149.83**	**276.67**	**574.00**	**71.75%**	—

2. Formulas Applied in Different Cells

The following standard spreadsheet formulas were used for row 2 (Arvind) and replicated across subsequent rows (Row 2 to Row 7):

Total Marks Formula (Cell E2):

```
=SUM(B2:D2)
```

Percentage Formula (Cell F2):

```
=(E2 / 800) * 100 [or =E2/800 formatted as Percentage]
```

Division Formula (Cell G2 - Nested IF Condition):

```
=IF(F2>=75, "1st with Distinction", IF(F2>=60, "1st Division", IF(F2>=50, "2nd Division", IF
(F2>=35, "3rd Division", "Fail"))))
```

3. Sequence of Steps Performed to Create the Worksheet

To construct the worksheet, apply formatting, and implement automated formula calculations in spreadsheet software (e.g., Microsoft Excel or LibreOffice Calc), the following steps were performed:

1. **Worksheet Creation & Data Entry:** Opened a fresh workbook sheet and entered column labels across row 1: `A1: Student Name`, `B1: Marks in Hindi out of 200`, `C1: Marks in Maths out of 200`, `D1: Marks in Finance out of 400`, `E1: Total (out of 800) (T)`, `F1: Percentage of Marks (P)`, and `G1: Division (D)`.
2. **Entering Raw Data:** Entered student names from `A2:A7` and their corresponding subject marks in columns B, C, and D according to the problem table.
3. **Calculating Total Marks (T):** Selected cell E2 and entered the formula `=SUM(B2:D2)`. Used the *Fill Handle* tool (dragged cell pointer down from E2 to E7) to instantly compute total marks for all students.
4. **Calculating Percentage (P):** Selected cell F2 and typed `=E2/800`. Applied *Percentage Formatting* with 2 decimal places from the toolbar, or entered `=(E2/800)*100` to yield standard numeric percentages. Auto-filled the formula down to cell F7.
5. **Constructing the Nested IF Formula for Division (D):** Clicked cell G2 and formulated the conditional hierarchy:
   `=IF(F2>=75, "1st with Distinction", IF(F2>=60, "1st Division", IF(F2>=50, "2nd Division", IF(F2>=35, "3rd Division", "Fail"))))`
6. **Replicating Formulae Across All Rows:** Dragged the fill handle from G2 down through G7. The relative cell references automatically adjusted to evaluate each student's respective percentage.
7. **Formatting & Final Review:** Applied cell borders, centered numeric columns, adjusted column widths for text clarity, and calculated class averages in row 8 using `=AVERAGE(B2:B7)`.

4. Verification Matrix of Student Results

Below is a summary verifying the individual calculations against the required division criteria:

Student	Total (T)	Percentage (P)	Condition Satisfied	Resulting Division (D)
Arvind	600 / 800	75.00%	P >= 75	1st with Distinction
Ajay	630 / 800	78.75%	P >= 75	1st with Distinction
Rehman	546 / 800	68.25%	60 <= P < 75	1st Division
Roger	484 / 800	60.50%	60 <= P < 75	1st Division
Samay	539 / 800	67.38%	60 <= P < 75	1st Division
Baljeet	645 / 800	80.63%	P >= 75	1st with Distinction